

Generating Buildable Areas and Fitting Building Footprints for Laneway Houses

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1 Tools, Methodology, and Techniques

Determining whether a laneway house can be legally built on a lot is a geospatial problem. All geospatial analysis was completed using Python scripts in combination with QGIS, a free GIS software. One Python script finds buildable areas where laneway houses can legally be built, and another determines if a sample building footprint fits within those buildable areas. These are generic scripts that can run for multiple municipalities, including Burnaby and Calgary, despite differences in GIS schema. To implement the solution for a municipality, only a list of parameters needs to be provided that outlines extracted rules for that municipality along with schema details.

1.1 Labelling Lot Boundaries

The first step to obtaining buildable area is labelling each lot boundary as front, side, or rear. Though most lots are simple rectangles, there are many irregular shapes as well. A generic algorithm was developed that can label lot boundaries for lots of any shape, outlined in Figure 1.

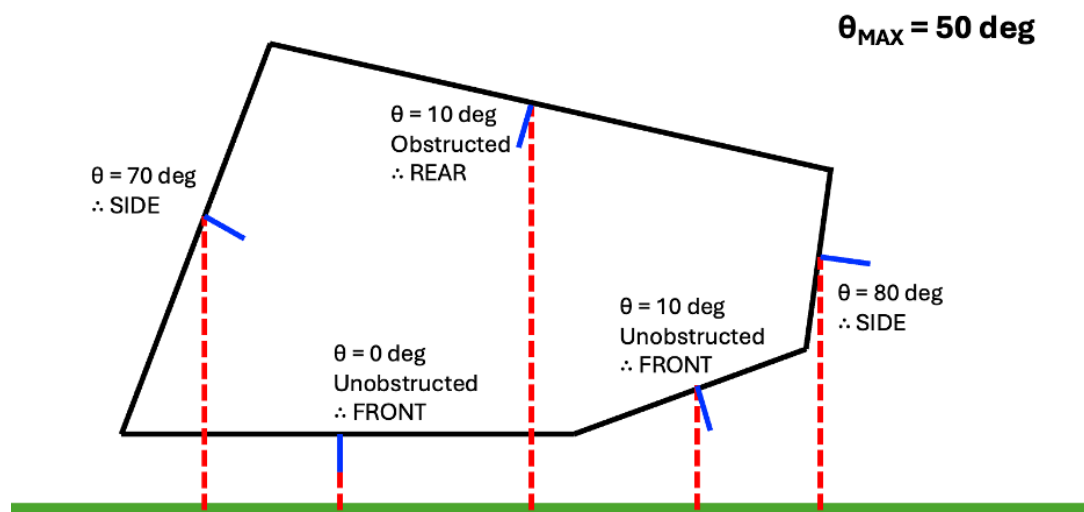


Figure 1: Diagram to outline the lot boundary labelling algorithm. Bee-lines (red) extend from the lot boundary midpoints to the street (green). The angle, θ , between a bee-line and the perpendicular bisector (blue) plays an important role in labelling.

First, the algorithm finds every lot boundary's bee-line, the shortest line between the boundary's midpoint and its street. Then, it finds the acute angle between each bee-line and the boundary's perpendicular bisector. This angle, θ , represents how much the boundary faces its street. If θ is below θ_{MAX} , then it is either a front or rear boundary. If the bee-line does not intersect any other boundary, then it is a front boundary. If it intersects a boundary from the same lot, then it is a rear boundary. Otherwise, it is a side boundary. A value of $\theta_{MAX} = 50$ deg is used so that boundaries with $\theta = \sim 45$ deg are labelled consistently even with minor deviations in angle.

1.2 Applying Setbacks

Setback rules determine how far a laneway house needs to be from different lot boundaries. The particular setback rule that is applied depends on the boundary's label and whether the boundary borders a neighbouring lot, street, or lane (outlined in Table 1). By subtracting the illegal spaces to build along the boundaries, the remaining space is the buildable area if there were no existing structures.

Boundary Label	Boundary Type	Setback Applied
Front	Street	Street Non-Flanking
Side	Neighbouring lot	Interior Side
	Street	Street Flanking
	Lane	Interior Side
Rear	Neighbouring lot	Interior Rear
	Street	Street Flanking
	Lane	Lane
Table 1: Conditions under which different setbacks are applied.		

1.3 Subtracting Existing Structures

For the purpose of developing laneway houses, only principal buildings need to be considered. Accessory buildings such as detached garages and sheds can be demolished. The largest building on each lot is labelled as principal, others are labelled as accessory. Principal buildings are subtracted from buildable areas with a buffer to account for the minimum distance allowed between front and rear principal buildings. The remaining space is the true buildable area where a laneway house can legally be built.

1.4 Fitting a Laneway House Footprint

To determine if a building's footprint fits in a lot's buildable area (Figure 2), many positions and rotations are tested. First, starting positions are placed in a grid spaced every 4 m that fall inside the lot. From each starting position, a greedy search repeatedly moves the footprint in whichever of eight directions that reduces the footprint area beyond the buildable area most. The translational step begins at 0.5 m and halves every time no direction improves the fit. The search stops once the footprint either becomes fully contained or can no longer reduce its area beyond the buildable area. Each starting position is tested using every possible footprint rotation in 10 deg steps until containment is found. Since the scope of the project is limited to laneway houses, placement is only allowed if the footprint's centroid is closer to the lot's average rear centroid than its average front centroid. In other words, laneway houses must be closer to the rear than the front.

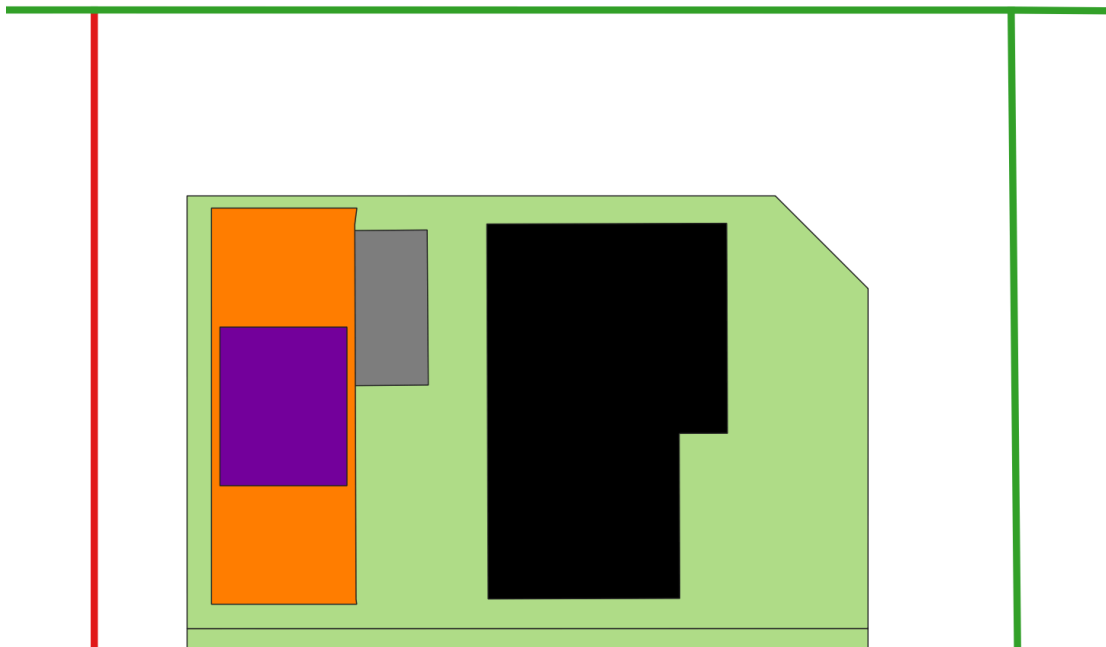


Figure 2: Lot (light green) with a fitted footprint (purple) within the buildable area (orange). Buildable area accounts for the principal building (black), but ignores the accessory building (grey). The lot borders two streets (dark green) and a lane (red).

2 Analysis and Interpretation

Upon visual inspection, the GIS solutions to define buildable areas and fit footprints are extremely accurate and robust to irregularly shaped lots. Unfortunately, there are no GIS tests to calculate accuracy. With more time and resources, a professional surveyor could be hired to manually identify the correct setback to apply to lot boundaries across a subset of lots in different municipalities. Then, the result of the scripts can be compared with the surveyor's verdict to calculate the percentage of lot boundaries that apply the correct setback. Similarly, buildings could be manually labelled to calculate the accuracy of that component of the algorithm.

Despite lacking GIS tests, cul-de-sacs were noticed to sometimes have front boundaries mislabelled as sides. This is because streets are represented as centrelines instead of polygons, so front bee-lines can be erroneously oblique, as seen in Figure 3. This could be addressed by developing an algorithm to convert street centrelines to representative polygons.

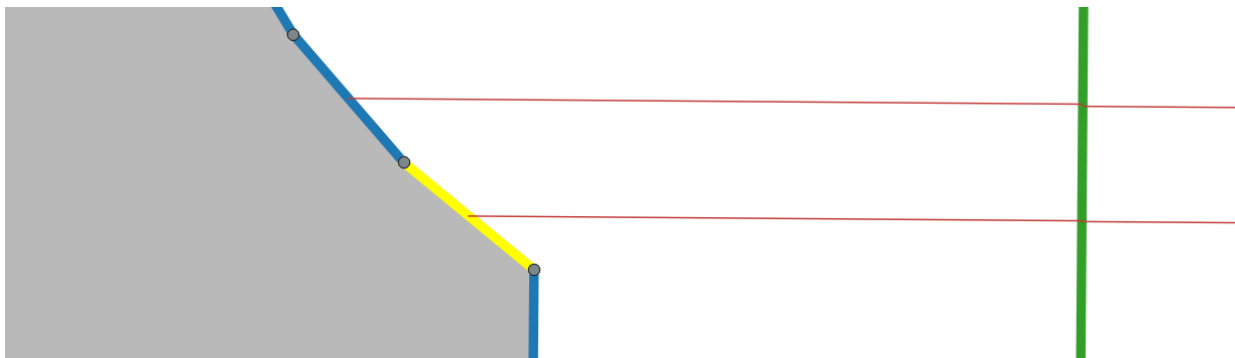


Figure 3: Zoomed-in on a lot in a cul-de-sac (grey). Bee-lines (red) extend from the lot boundary midpoints to the street centreline (green). The yellow boundary is erroneously identified as a side instead of a front because it doesn't sufficiently face the street's centreline, even though it does in fact face the street.

Mislabelled lot boundaries can also occur when a single street wraps around a lot. In these cases, the closest point from a rear boundary midpoint to the street can be at such a great angle that the rear is misidentified as a side, as seen in Figure 4.

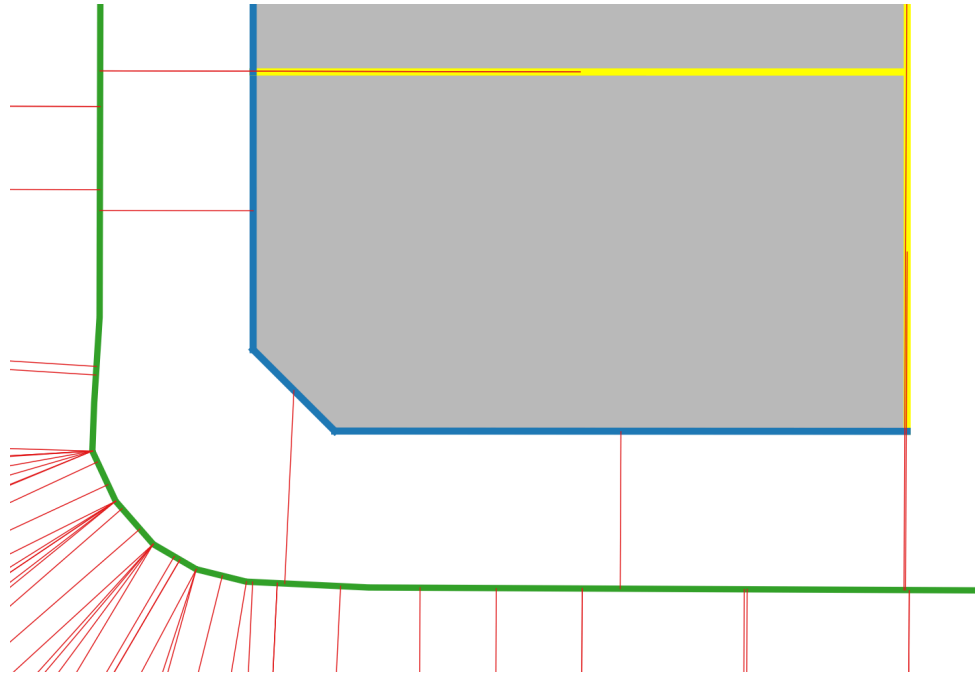


Figure 4: Lot (grey) with a rear boundary mislabelled as a side on the righthand side due to a wrapping street. Bee-lines (red) extend from the lot boundary midpoints to the street (green).

3 Conclusion

The GIS algorithms used to generate buildable areas and fit footprints are extremely accurate and robust to irregularly shaped lots based on a visual inspection. However, boundaries are sometimes mislabelled in cul-de-sacs and when a street wraps around a lot. There were insufficient resources to develop GIS tests to calculate accuracy. However, this could be done in the future with the help of a professional surveyor to obtain a reference with guaranteed fidelity. With tests, parameters in the algorithm such as θ_{MAX} could be fine-tuned to maximize accuracy.

When subtracting existing structures from buildable areas, the current algorithm does not distinguish between existing laneway houses and accessory buildings. Both are ignored when defining buildable areas. This is a problem since housing developers do not typically tear down one laneway house to build another. If there was more time, methods to distinguish between existing laneway houses and accessory buildings would have been explored. It may be impossible to reliably determine from footprint geometry and position alone.

Although the GIS scripts can support the schemas for Burnaby and Calgary, there is not yet a generic solution to support the schema of any potential municipality. Having conditions in the scripts themselves to handle different schemas, as is done currently, is not a scalable solution. Ideally, an automated solution can be developed to convert required GIS files to a standardized schema to allow for simple scalable scripts.